From interval palettes to dense nonsumsets in prime fields

Project proof artifact for AIM Problem 2.7

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Abstract

For a prime p, let $M(p)$ be the greatest cardinality of a subset of $\mathbb{F}_p$ which is not $B+B$ for any $B \subseteq \mathbb{F}_p$, with equal summands allowed. We prove

$$M(p) = p - p^{1/2+o(1)}.$$

The new deduction combines Noga Alon's universal large-set theorem with the interval-palette theorem in Shouqiao Wang's pinned proposed solution of Erdős Problem 788. Two interval palettes are transferred to consecutive charts of $\mathbb{Z}_p$; an exact finite counting argument then chooses a second deletion set which cannot be realized by any self-sumset root. We state the imported theorem and all reductions explicitly. Wang's source is machine-checked and was rebuilt locally, but is recent and not peer-reviewed; this artifact therefore distinguishes mathematical deduction from community acceptance and novelty.

1 Statement and notation

For $B \subseteq \mathbb{F}_p$, define the unrestricted self-sumset

$$B + B = \{b_1 + b_2 : b_1, b_2 \in B\}.$$

The empty root is allowed, and $b_1 = b_2$ is allowed. Define

$$M(p) = \max\{|A| : A \subseteq \mathbb{F}_p \text{ and } A \neq B + B \text{ for every } B \subseteq \mathbb{F}_p\}$$

and put $d(p) = p - M(p)$. Thus $d(p)$ is the strict complement threshold, not the inclusive threshold used in some earlier papers.

Theorem 1 (AIM Problem 2.7)**.** *As p tends to infinity through the primes,*

$$d(p) = p^{1/2+o(1)} \qquad \text{and} \qquad M(p) = p - p^{1/2+o(1)}.$$

All logarithms are natural. We identify $\mathbb{F}_p$ with the additive group $\mathbb{Z}_p$ and with its standard representatives $\{0, \ldots, p-1\}$.

2 Two imported results

The lower estimate follows from the following exact inclusive statement.

Theorem 2 (Alon, Theorem 2.1 [2])**.** *For every sufficiently large prime p, if $F \subseteq \mathbb{Z}_p$ and*

$$|F| \le \frac{1}{4000}\sqrt{\frac{p}{\log p}},$$

then $\mathbb{Z}_p \setminus F = A + A$ for some $A \subseteq \mathbb{Z}_p$.

We next state exactly the consequence of Wang's interval theorem which is used here. For $N \geq 2$ and $Q \subseteq \{1, \ldots, 2N-3\}$, let H_Q be the simple graph on

$$[N]_0 = \{0, 1, \ldots, N-1\}$$

in which distinct x, y are adjacent precisely when $x + y \in Q$.

Theorem 3 (Interval-palette theorem [3]). *There is an absolute constant $C > 0$ such that, for every sufficiently large integer N, there is $Q_N \subseteq \{1, \ldots, 2N-3\}$ satisfying*

$$|Q_N| + \alpha(H_{Q_N}) \leq (N+1)^{1/2+\Delta_N}, \qquad \Delta_N = C\left(\frac{\log\log(N+1)}{\log(N+1)}\right)^{1/3}.$$

For completeness, this is the exact normalization of the cited result. Wang writes

$$I_n = (n, 2n) \cap \mathbb{N} = \{n+1, \ldots, 2n-1\}$$

and proves both the min–max identity

$$f(n) = \min_{D \subseteq J_n} \big(|D| + \alpha(G_D)\big),$$

with the minimum unchanged when D is restricted to attainable sums of two distinct elements of I_n, and the upper bound

$$f(n) \leq n^{1/2+C(\log\log n/\log n)^{1/3}}.$$

Set $n = N+1$, translate I_n by $-(n+1)$, and translate every palette sum by $-(2n+2)$. The vertex set becomes $[N]_0$, the attainable distinct sums become exactly $\{1, \ldots, 2N-3\}$, and the graph and its independence number are preserved. This gives Theorem 3 without an asymptotic change of parameters.

3 Transfer to a cyclic palette

For $E \subseteq \mathbb{Z}_p$, let Γ_E be the simple graph on $\mathbb{Z}_p$ in which distinct x, y are adjacent exactly when $x + y \in E$.

Lemma 4 (Two-chart transfer). *For every sufficiently large odd prime p, there is $E \subseteq \mathbb{Z}_p$ such that, with*

$$e = |E|, \qquad r = \alpha(\Gamma_E), \qquad H = e + r,$$

one has

$$2\sqrt{p} - 1 \leq H \leq p^{1/2+\eta_p}$$

for a sequence $\eta_p \to 0$.

Proof. Put

$$N_0 = \frac{p+1}{2}, \quad N_1 = \frac{p-1}{2}, \quad a_0 = 0, \quad a_1 = N_0,$$

and partition $\mathbb{Z}_p$ into the two standard intervals

$$V_i = a_i + \{0, \ldots, N_i - 1\} \qquad (i = 0, 1).$$

Choose Q_{N_i} from Theorem 3 and define

$$E_i = \{2a_i + q \pmod{p} : q \in Q_{N_i}\}, \qquad E = E_0 \cup E_1.$$

Because $Q_{N_0} \subseteq \{1, \dots, p-2\}$ and $Q_{N_1} \subseteq \{1, \dots, p-4\}$, reduction modulo p is injective on each palette. Cross-palette collisions are allowed and only decrease $|E|$. Thus

$$e \leq |Q_{N_0}| + |Q_{N_1}|. \tag{3.1}$$

Let R be independent in Γ_E. For fixed i, define

$$R_i = \{u \in [N_i]_0 : a_i + u \in R\}.$$

If distinct $u, v \in R_i$ satisfied $u + v \in Q_{N_i}$, then

$$(a_i + u) + (a_i + v) \equiv 2a_i + u + v \pmod{p} \in E_i \subseteq E,$$

contradicting independence. Hence R_i is independent in $H_{Q_{N_i}}$. The two charts partition $\mathbb{Z}_p$, so

$$r \leq \alpha(H_{Q_{N_0}}) + \alpha(H_{Q_{N_1}}). \tag{3.2}$$

Let $\varepsilon_p = \max_i \Delta_{N_i}$ and $\eta_p = \varepsilon_p + \log 2/\log p$. Equations (3.1), (3.2), and Theorem 3 give

$$H \leq \sum_{i=0}^{1} (N_i + 1)^{1/2+\Delta_{N_i}} \leq 2p^{1/2+\varepsilon_p} = p^{1/2+\eta_p}.$$

Since $N_i \to \infty$, we have $\eta_p \to 0$.

For the lower estimate, each vertex of Γ_E has degree at most e: for fixed x and fixed $s \in E$, at most one neighbor can satisfy $x + y = s$. A graph of maximum degree e has an independent set of size at least $p/(e+1)$ by greedy coloring. Therefore

$$H = e + r \geq e + \frac{p}{e+1} = (e+1) + \frac{p}{e+1} - 1 \geq 2\sqrt{p} - 1.$$

This argument uses only distinct-pair edges, so no diagonal convention has been changed. □

4 The exact-output deletion

Lemma 5 (Strict finite counting gap)**.** *With e, r, H, η_p as in Lemma 4, for every sufficiently large p,*

$$3 \leq H, \qquad 4H \leq p - e,$$

and

$$\binom{p-e}{4H} > \sum_{j=0}^{r} \binom{p}{j}. \tag{4.1}$$

Proof. Take p large enough that

$$\eta_p \leq \frac{1}{16}, \qquad \frac{\log 8}{\log p} \leq \frac{1}{16}, \qquad 5p^{9/16} \leq p. \tag{4.2}$$

Lemma 4 gives

$$2\sqrt{p} - 1 \leq H \leq p^{1/2+\eta_p} \leq p^{9/16}.$$

Hence $H \geq 3$ eventually and, because $e \leq H$, $e + 4H \leq 5p^{9/16} \leq p$.

For integers $0 < k \leq n$,

$$\binom{n}{k} = \prod_{i=0}^{k-1} \frac{n-i}{k-i} \geq \left(\frac{n}{k}\right)^k,$$

because $(n-i)/(k-i) \geq n/k$ is equivalent to $i(n-k) \geq 0$. Apply this with $n = p - e$ and $k = 4H$. Since $e \leq H \leq p^{9/16} \leq p/2$ eventually,

$$\log \binom{p-e}{4H} \geq 4H \log \frac{p-e}{4H} \geq 4H \log \frac{p}{8H}.$$

Using (4.2) and $H \leq p^{9/16}$, the last logarithm is at least $\frac{3}{8} \log p$, and therefore

$$\binom{p-e}{4H} \geq p^{3H/2}. \tag{4.3}$$

On the other hand,

$$\sum_{j=0}^{r} \binom{p}{j} \leq \sum_{j=0}^{r} p^j \leq p^{r+1} \leq p^{H+1} \leq p^{4H/3} < p^{3H/2},$$

where $H + 1 \leq 4H/3$ follows from $H \geq 3$. Combining this strict inequality with (4.3) proves (4.1). $\square$

Lemma 6 (Unrealized output)**.** *For every sufficiently large prime p, there is $T \subseteq \mathbb{Z}_p \setminus E$ with $|T| = 4H$ such that*

$$A_p = \mathbb{Z}_p \setminus (E \cup T)$$

is not $B + B$ for any $B \subseteq \mathbb{Z}_p$. Moreover,

$$|\mathbb{Z}_p \setminus A_p| \leq 5p^{1/2+\eta_p}.$$

Proof. Let $\mathcal{T}$ be the family of all $4H$-element subsets of $\mathbb{Z}_p \setminus E$. By Lemma 5,

$$|\mathcal{T}| = \binom{p-e}{4H} > \sum_{j=0}^{r} \binom{p}{j}. \tag{4.4}$$

Call $T \in \mathcal{T}$ realized if

$$B + B = \mathbb{Z}_p \setminus (E \cup T) \tag{4.5}$$

for some $B \subseteq \mathbb{Z}_p$. Equation (4.5) implies $(B+B) \cap E = \varnothing$. Thus no distinct pair $x, y \in B$ has $x + y \in E$, so B is independent in Γ_E and $|B| \leq r$. There are at most the right-hand side of (4.4) such roots. In addition, a fixed root determines at most one deletion set, because (4.5) forces

$$T = (\mathbb{Z}_p \setminus (B+B)) \setminus E. \tag{4.6}$$

Consequently (4.4) leaves some T unrealized. The corresponding A_p is a nonsumset. Finally, $E \cap T = \varnothing$ and $e \leq H$, so

$$|\mathbb{Z}_p \setminus A_p| = e + 4H \leq 5H \leq 5p^{1/2+\eta_p}.$$

Equal summands cause no gap: they remain part of (4.5), while distinct pairs provide the necessary independence condition and (4.6) uses the full sumset. $\square$

5 Proof of the main theorem

Proof of Theorem 1. Set

$$q_p = \left\lfloor \frac{1}{4000}\sqrt{\frac{p}{\log p}} \right\rfloor.$$

If $S \subseteq \mathbb{Z}_p$ has $|S| \geq p - q_p$, then its complement satisfies the inclusive hypothesis of Theorem 2; hence S is a self-sumset. No nonsumset has size at least $p - q_p$, and integrality gives

$$M(p) \leq p - q_p - 1, \qquad d(p) \geq q_p + 1 > \frac{1}{4000}\sqrt{\frac{p}{\log p}}. \tag{5.1}$$

Lemma 6 supplies a nonsumset A_p with complement at most $5p^{1/2+\eta_p}$. Hence

$$d(p) = p - M(p) \leq p - |A_p| \leq 5p^{1/2+\eta_p}. \tag{5.2}$$

Taking logarithms in (5.1) and (5.2) yields

$$\frac{1}{2} - \frac{\log\log p}{2\log p} - \frac{\log 4000}{\log p} < \frac{\log d(p)}{\log p} \leq \frac{1}{2} + \eta_p + \frac{\log 5}{\log p}.$$

Both outer expressions tend to 1/2. Therefore $d(p) = p^{1/2+o(1)}$, and the defining identity $M(p) = p - d(p)$ gives $M(p) = p - p^{1/2+o(1)}$. $\square$

6 Verification and evidence boundary

The imported Wang repository was pinned at commit

`d28713ac8245ca86a686b8c67370a8d19d81b242`.

Its Lean 4.27 development used Mathlib commit

`a3a10db0e9d66acbebf76c5e6a135066525ac900`

and completed all 7,936 local build jobs. The final theorem depends only on `propext`, `Classical.choice`, and `Quot.sound`; the audited source contains no `sorry`, `axiom`, or `admit`. The interval normalization above was checked against both the paper and the formal definitions.

The new cyclic transfer and deletion proof passed a separate QED xhigh structural audit, a detailed step-by-step audit, deterministic finite falsification checks, and final verdict `DONE`. These checks establish the repository's proof claim relative to the pinned imported theorem. They do not establish novelty, peer review, or community acceptance. In particular, the source itself is titled a proposed solution and the public Erdős problem page remained marked open at the audit date.

References

[1] E. Croot and V. F. Lev (collectors), *Problems Presented at the Workshop on Recent Trends in Additive Combinatorics*, AIM workshop list, Problem 2.7 (B. Green), 2004. `https://aimath.org/WWN/additivecomb/additivecomb.pdf`.

[2] N. Alon, *Large sets in finite fields are sumsets*, J. Number Theory 126 (2007), 110–118. `https://web.math.princeton.edu/~nalon/PDFS/sumset.pdf`.

[3] S. Wang, *A Proposed Solution to Erdős Problem 788*, pinned repository artifact, commit d28713ac8245ca86a686b8c67370a8d19d81b242, 2026. https://github.com/ShouqiaoW/erdos/tree/d28713ac8245ca86a686b8c67370a8d19d81b242/788.